

Oscar Giraldo Castillo

Senior Full-Stack Engineer

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Summary

Senior full-stack engineer with 14+ years shipping production backends and AI-native systems across fintech, edtech, agtech, medtech, and SaaS. Works end-to-end, from Python / Django, FastAPI, and Next.js services on AWS, to Rust APIs, LLM tooling (MCP, RAG, multi-provider LLMs), event-driven architectures, and ESP32 / ESP-IDF embedded firmware. Strong in distributed systems, observability and product ownership. Previously co-founded Chaty and led engineering at Artificio.

Experience

Senior Fullstack Developer

Oct 2024 - Present

Cometa

Remote (Ciudad de México, México)

- Member of Cometa's **finance engineering team**. Own payins, payouts, and billing / invoicing flows end-to-end across a Django modular monolith and Next.js frontend.
- Built **Argus**, a Rust read-only Redshift proxy (Axum, lambda_http on AWS Lambda + API Gateway) powering the Guardian Credit Scoring flow: whitelisted tables, identity-based row filtering, pagination, and thread-safe credential refresh.
- Delivered the billing + inventory domain: cart payment lifecycle state machine, batch-only inventory reservation pipeline, cart-expiration via cross-module outbox events, Gandalf-authenticated endpoints, and Postgres indexes on hot paths.
- Shipped Guardian Stripe checkout end-to-end (PaymentIntent, per-currency minimum-amount drawer, locale-aware errors, ally) across Django backend and Next.js frontend.
- Built MCP servers exposing Cometa's domain APIs as LLM-callable tools; shipped full-stack features on a Django modular monolith + Next.js (Clean Architecture, DDD) with on-call triage, Alembic migrations, and report-query tuning.

Senior Software Engineer

Sep 2023 - Oct 2024

Foris

Remote

- Built Airflow DAGs and an in-house DSL for data pipelines on AWS Glue, S3, and Athena.
- Shipped Python backend services for academic planning and course scheduling products; led technical design reviews and mentored mid-level engineers.

Director of Software Engineering

Jan 2023 - Sep 2023

Artificio

Remote

- Shipped an end-to-end image-search service for **Seezlo** (artwork discovery): deployed vision models on AWS SageMaker and an image-embedding RAG for artwork retrieval, exposed via FastAPI.
- Led the engineering function end-to-end: hiring, roadmap, and architecture direction for Python services on AWS.

Python Backend Engineer

Oct 2021 - Jan 2023

SpaceAG Global

Lima, Perú

- Implemented a multi-tenant reporting module syncing AWS RDS with AWS Athena via EventBridge, Celery, and Pandas, producing Parquet files on S3.
- Scaled the Postgres master on Aurora Serverless v2 with read replicas; built AWS Lambda health-checks wired into Slack, Sentry, and Google Sheets.
- Shipped Django/Flask APIs on AWS Lambda (Serverless, Zappa); exposed a legacy Microsoft Access DB through API Gateway + DynamoDB and integrated weather stations (WeatherLink v2, Metos 2.0, Adcon).

Backend Developer & DevOps

Jan 2020 - Dec 2021

Alfi

Remote

- Rebuilt the core application from scratch in Django and modeled all REST APIs; production stack on AWS ECS/Fargate, RDS, ElastiCache (Redis), and RabbitMQ.
- Led the backend team: story points, code reviews, and PRs.

Co-founder & Backend Engineer

Jan 2017 - Jan 2020

Chaty

Lima, Perú

- Co-founded a multi-tenant SaaS for restaurant-chain chat ordering; integrated Facebook Messenger, WhatsApp, and Telegram APIs for chatbot-driven order flows.
- Integrated multiple payment gateways for end-to-end checkout and automated shipping assignment + order routing on PostGIS.

Software Architect

Oct 2014 - Jan 2017

Ventorystack

Remote

- Designed and implemented a real-time inventory system on Tornado, MongoDB, AWS Lambda, and AWS SQS.

Earlier roles: Django edtech, embedded systems, ML research

2011 - 2015

Crehana (CTO), DGNES (Founder), Engitronic, CTIC-UNI

Lima, Perú

Projects

MCP Servers for Cometa | *Python, Model Context Protocol, Claude, OpenAI*

- Open-source and internal MCP server implementations exposing Cometa domain APIs (payments, admissions, communications) as LLM-callable tools for AI developer tooling.

Chapatuplaza MCQ generator | *Python, LangChain, Gemini 2.5, Claude, Qdrant, AWS Lambda, SQS*

- Event-driven RAG for Peruvian medical-exam prep (SERUMS/MINSA): vision-based PDF OCR and image-embedding retrieval on Qdrant; multi-provider LLM generation (Gemini with Claude fallback) on AWS Lambda, with async SQS callbacks (DLQ-backed) for persistence and Langfuse observability.

Cometag NFC reader firmware | *C, ESP-IDF, XIAO ESP32-S3, PN532, I2C*

- Embedded firmware for a XIAO ESP32-S3 driving a PN532 NFC reader over I2C; detects ISO14443A tags (MIFARE, NTAG) and streams UIDs over serial USB to prototype NFC workflows at Cometa.

Fabrimetal | *Next.js, DRF, PostgreSQL, Docker, AWS*

- Full-stack e-commerce for laser-cutting and metal-bending services: Next.js 15 storefront, DRF backend with clean-architecture apps (controllers / services / repositories), explicit order state machine (draft, to-quote, pending-payment, paid, in-progress, delivered), Docker Compose, S3 storage, and Telegram-bot notifications for order workflow.

Skills

Languages: Python, Go, Rust, JavaScript/TypeScript, SQL, C/C++

Backend frameworks: Django, FastAPI, Flask, Django Ninja, Tornado, Axum (Rust)

Frontend: Next.js, React, TypeScript, Tailwind CSS

Databases: PostgreSQL, DynamoDB, MongoDB, AWS Athena, AWS Redshift, Qdrant, Pinecone

Messaging & async: Redis, RabbitMQ, AWS SQS, Celery, EventBridge, outbox pattern

Cloud & DevOps: AWS (Lambda, ECS, Fargate, S3, API Gateway, RDS, SageMaker), Docker, Kubernetes, Terraform, Serverless, CI/CD, Nginx

AI & ML: Model Context Protocol (MCP), LangChain, RAG, vector search, LLM providers (Claude, OpenAI, Gemini), AWS SageMaker, image embeddings, Pandas, TensorFlow

Embedded & IoT: ESP32 (ESP-IDF), Seed XIAO ESP32-S3, C firmware, PN532 NFC, I2C / SPI / UART, ISO14443A (MIFARE, NTAG), microcontrollers

Architecture & practices: Microservices, event-driven architecture, REST, GraphQL, Clean Architecture,DDD, BDD, TDD, SCRUM, Kanban

Observability: OpenTelemetry (OTEL Collector), Grafana, Loki, Tempo, Prometheus, Sentry, Langfuse

Operating systems: GNU/Linux (primary), macOS, Windows

Education

B.S. Mechatronics Engineering	2008 - 2013
<i>Universidad Nacional de Ingeniería (UNI)</i>	<i>Lima, Perú</i>
Activities: GISCIA, CEDIM, CTIC, python-uni. Relevant coursework: Matlab, algorithms.	
Marketing & Value Management (Summer course)	2016
<i>State University of New York at New Paltz</i>	<i>New Paltz, NY, USA</i>

Certifications

- **Curso de Go Intermedio: POO y Concurrencia**, Platzi (Dec 2025). platzi.com/p/oscargicast/curso/2357-golang-intermedio/diploma
- **Diploma de Especialización en IA, ML & Deep Learning con Python**, Universidad Nacional de Ingeniería, FIQT (Nov 2025), 120 hours, average grade 19/20.
- **Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning**, DeepLearning.AI (Jun 2024). coursera.org/verify/ZZMM346D2N7T
- **AWS Certified Cloud Practitioner**, Amazon Web Services (Mar 2024). credly.com/badges/9b39b6f3-2886-4465-b99c-b06017b88cde

Honors & Awards

- **Publication:** Development of a Web Database System for a Comprehensive Hypertension Management Program in the Primary Health Care Setting of Urban Areas of Peru. *Healthcare Informatics Research* (Jan 2019). doi.org/10.4258/hir.2019.25.1.41
- **Academic Awards:** IBS Brasil/SUNY New Paltz Executive Management (Nov 2016); UNI upper-third graduation.
- **2nd Place, Mechatronics Academic Olympics:** FESTO, Aug 2011 (pneumatic/PLC tests).

Languages

- **Spanish:** Native.
- **English:** Full Professional Proficiency; EF SET C2 Proficient (72/100). cert.efset.org/en/1xUWsa